

Real-time GS Mapping on Aria

Building useful maps under limited compute

박상원 , 김동휘 , 송채현

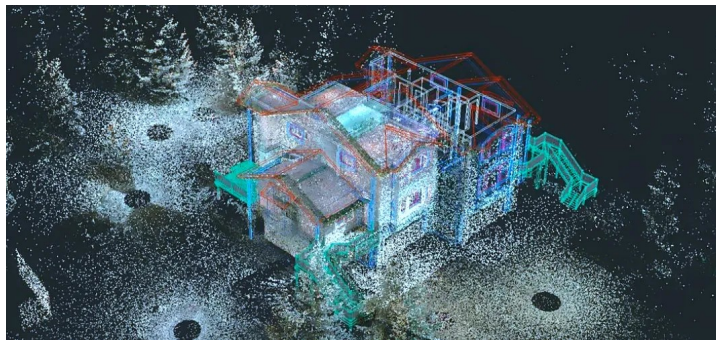
RPM Lab, Seoul National University

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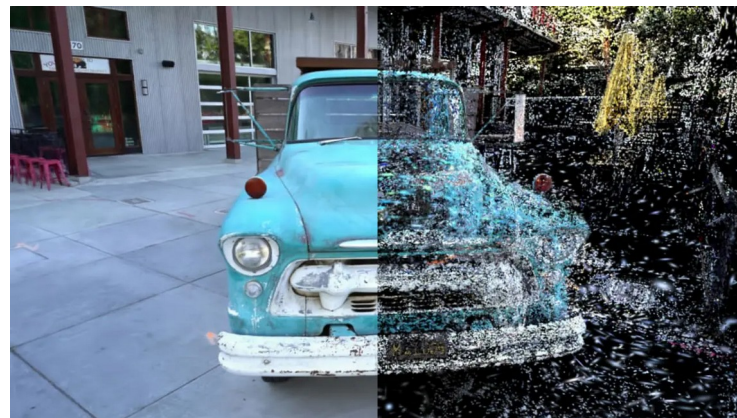
Why Gaussian Splatting?

Represent a scene with colored, semi-transparent 3D Gaussians.

Point cloud



Gaussian representation / rendering



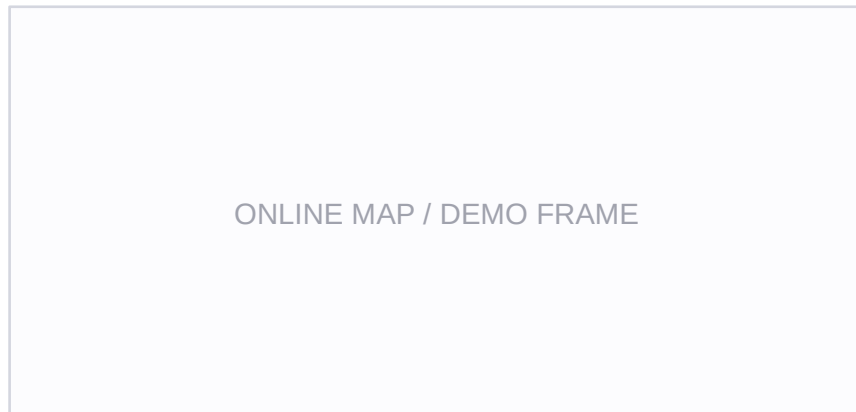
Photorealistic appearance + fast rendering

Why Real-time Mapping?

A robot or AR user needs a useful map while moving.



Observe the environment

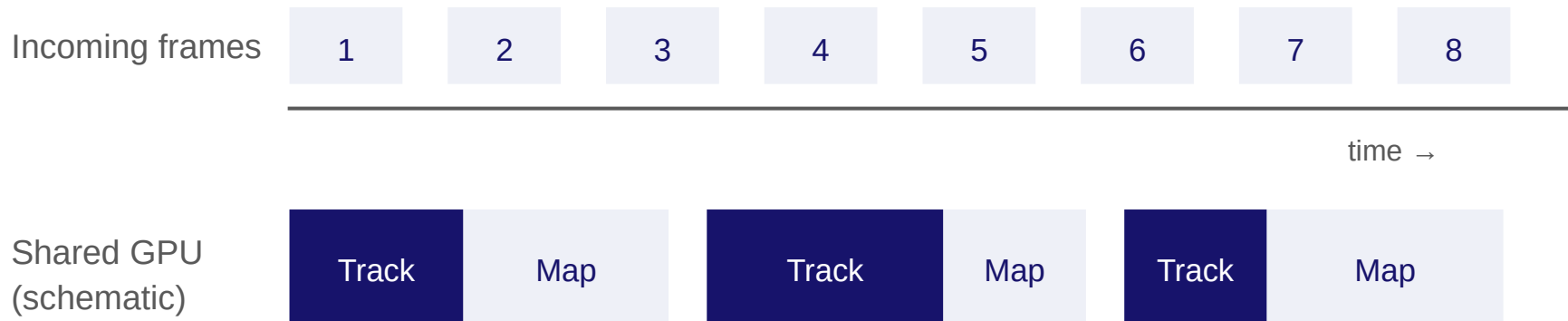


Build and use the map now

Fast rendering does not guarantee fast map convergence.

Mapping under Limited Compute

Tracking and mapping share a finite online compute budget.



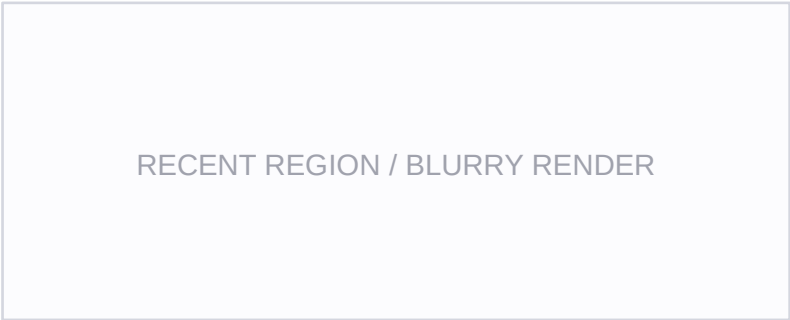
More mapping views compete for the same updates.

Use available observations within available computation.

What Makes Online Mapping Difficult?

Two requirements: convincing appearance and reliable geometry.

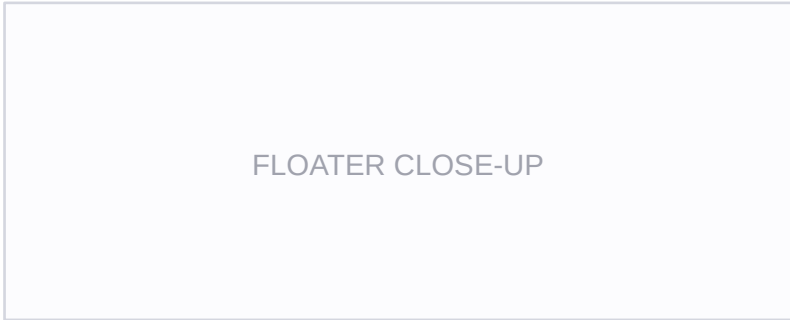
Under-optimized appearance



RECENT REGION / BLURRY RENDER

Too little useful supervision

Incorrect geometry



FLOATER CLOSE-UP

Opacity in observed free space

How quickly can a newly observed region become reliable?

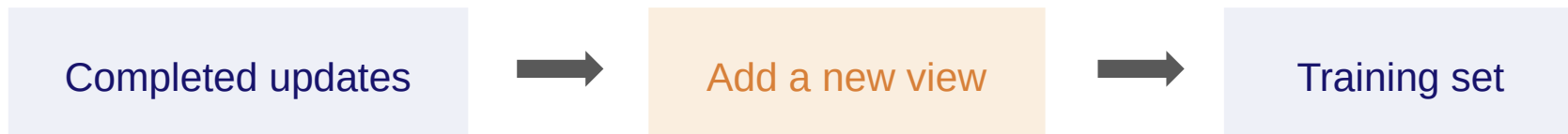
When Should We Add More Views?

Use frames between keyframes, paced by completed optimization.

Tracking
keyframes



Intermediate RGB frames also carry mapping information.

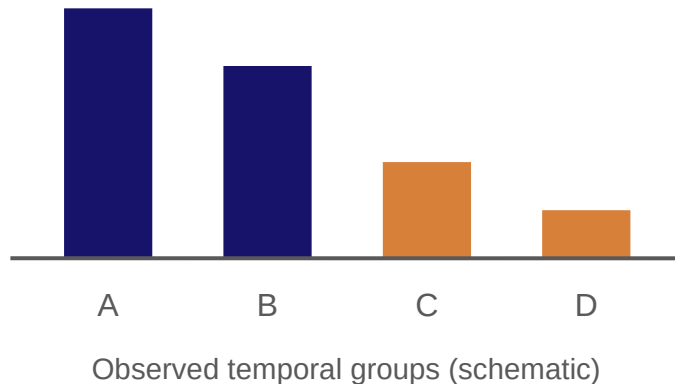


Grow supervision at the pace of optimization.

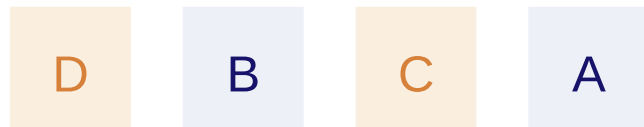
Which Views Should We Optimize?

Give under-trained observations more opportunities, while mixing views.

Uneven optimization exposure



Count-aware randomized selection

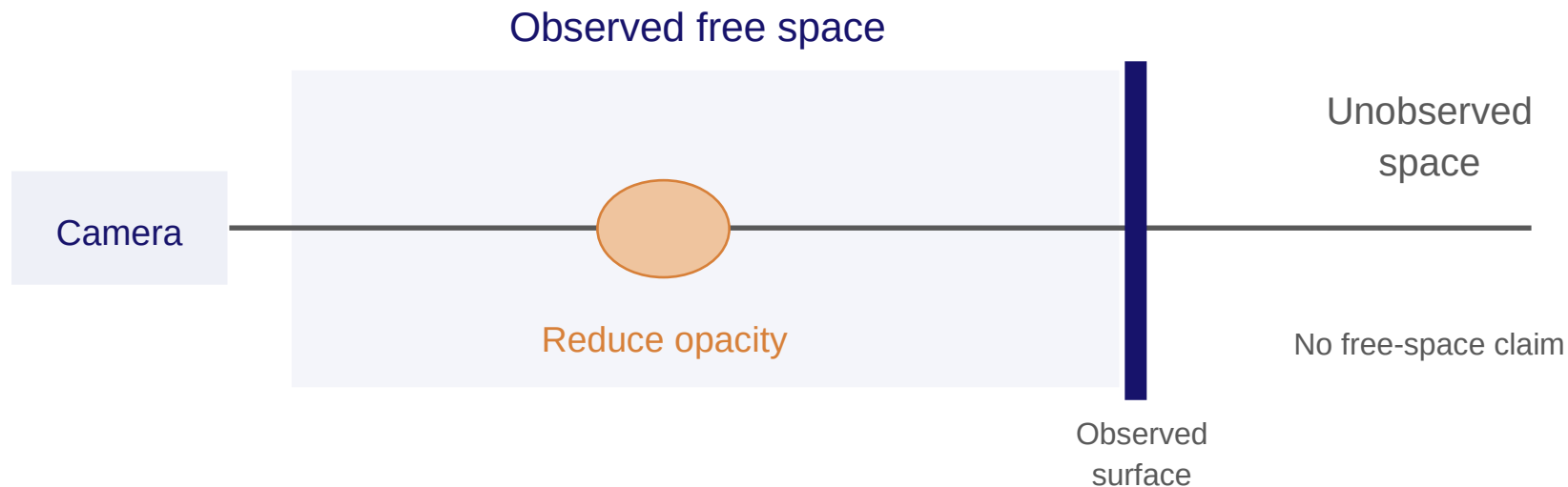


Correct severe deficits
Preserve sampling diversity

Balance learning opportunities without fixing a rigid order.

Keeping Observed Free Space Empty

A good color match can still hide an incorrect 3D structure.

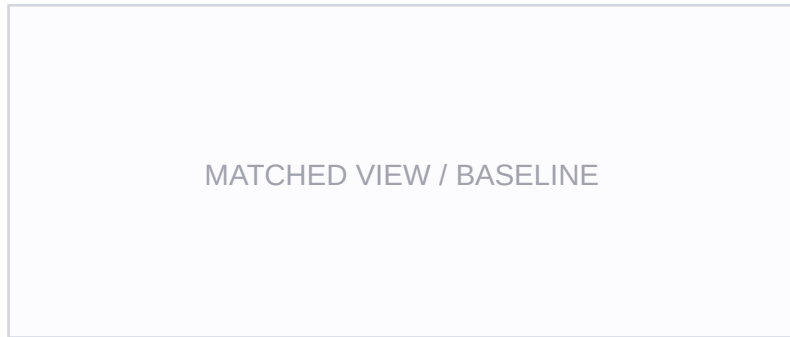


Use causal depth evidence to suppress premature opacity.

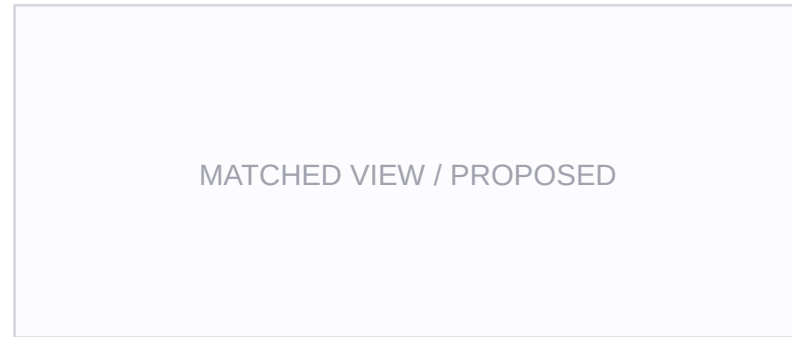
Current Progress & Evaluation

View growth and scheduling are implemented; joint validation is ongoing.

Baseline



Proposed setting



Next: matched-budget appearance + geometry evaluation

Takeaway & Next Step

Make each unit of online computation
produce a more useful map.

01

When to add

Pace view growth

02

What to optimize

Balance exposure

03

Where to constrain

Use free-space evidence

Next: validate appearance and geometry gains at the native input rate.